Dashboard

Risk Analysis

Predictive Analytics

Customer Segmentation

Suggestion

Advanced Credit Risk Analytics and Loan Default Prediction

Using Machine Learning , Data Mining and Predictive Analytics

Objective

To develop an end-to-end credit risk assessment system for identifying high-risk borrowers and predicting loan defaults using machine learning, customer segmentation, and interactive Power BI dashboards to support data-driven lending decisions.

Tools Used

- Python
- Pandas & NumPy
- Scikit-Learn
- XGBoost
- SHAP Explainable AI
- K-Means Clustering
- Association Rule Mining
- Power BI
- DAX
- GitHub

Machine Learning Models

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost *(Best Model)*

Dashboard

Executive Dashboard
- Portfolio health and lending overview

Risk Analytics Dashboard
- Default drivers and customer risk factors

Predictive Analytics Dashboard
- Model performance and feature importance

Customer Segmentation & Risk Profiling
- Risk clusters and borrower profiles

Recommendations, Conclusion & Future Scope

Prepared By

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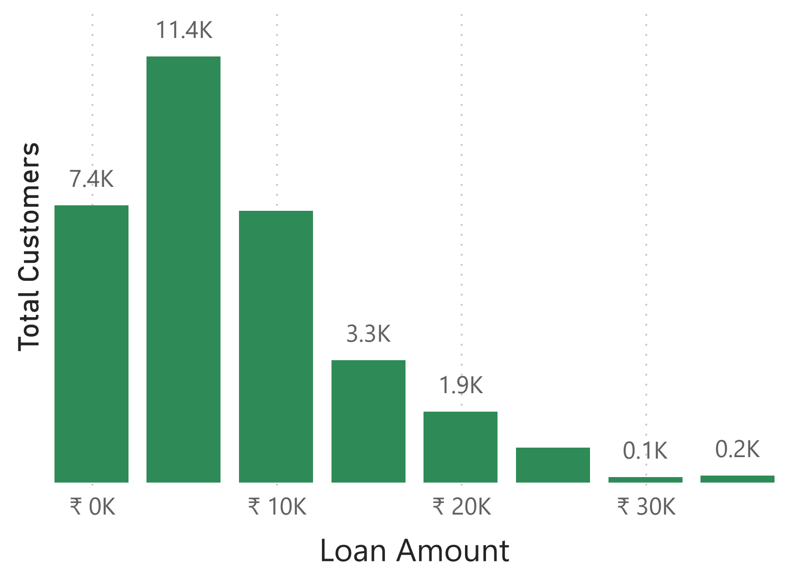
CREDIT RISK EXECUTIVE DASHBOARD

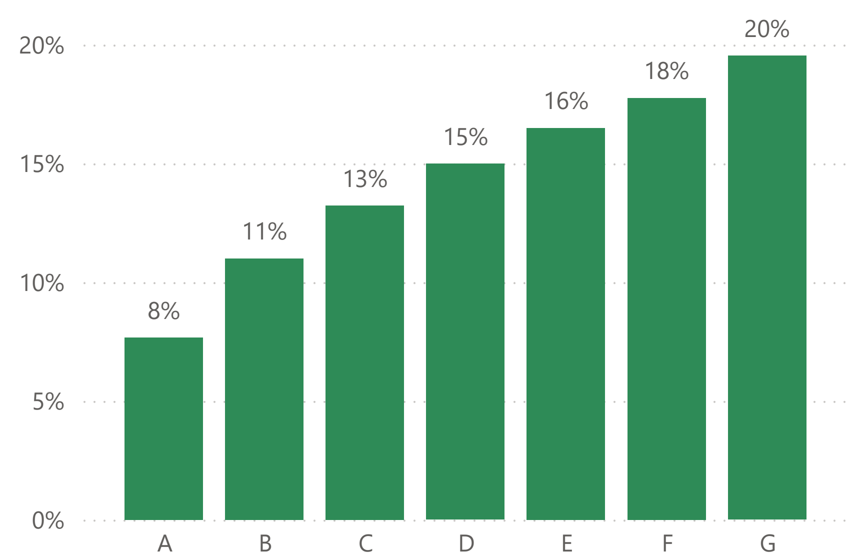
PORTFOLIO HEALTH AND LENDING OVERVIEW

Total Customers	Total Loan Amo...	Default Rate %	Total Defaults	Average Income	Average Interest...	Risk Status
32K	₹ 311.0M	21.87%	7K	66.09K	11%	High Risk

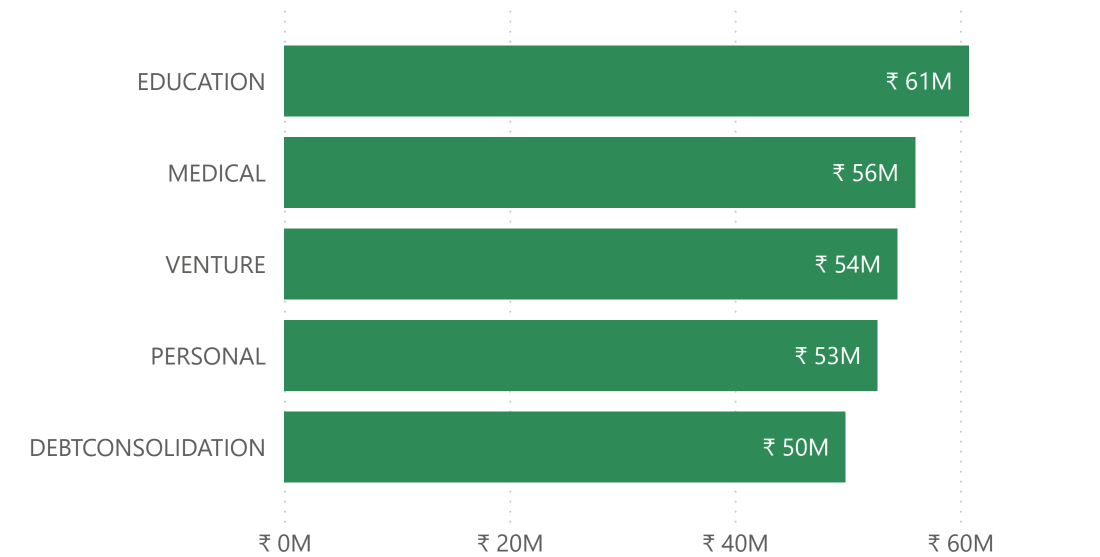
Loan Portfolio Distribution



Interest Rate by Loan Grade

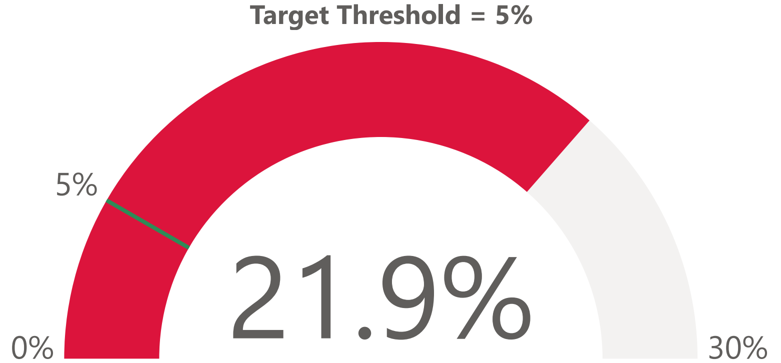


Top Loan Purpose

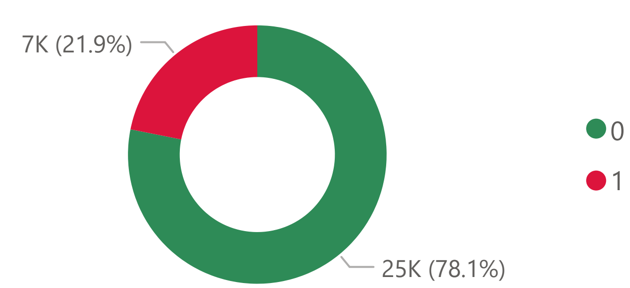


Portfolio Default Rate

Target Threshold = 5%



Portfolio Health



Key Insights

- The portfolio default rate is **21.9%**, significantly higher than the target threshold of **5%**, indicating a need for stronger credit risk management.
- **Grade G loans** have the highest average interest rate (**20%**), suggesting that lower credit-grade borrowers carry greater risk.
- **Education loans** represent the largest portion of the portfolio, contributing approximately **₹61 million**, followed by Medical and Venture loans.
- Portfolio health remains relatively stable, with **nearly 78% of borrowers repaying successfully**, while about **22% are in default**.
- Loan exposure is well diversified, with most customers concentrated in lower and medium loan amounts, reducing concentration risk.

Loan Purpose Filter

DEBTCONSOLIDATION	EDUCATION	HOMEIMPROVEMENT	MEDICAL	PERSONAL	VENTURE

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LOAN RISK ANALYSIS DASHBOARD

Default Rate by Loan Grade and Loan Purpose

Loan Grade	DEBTCONSOLIDATION	EDUCATION	HOMEIMPROVEMENT	MEDICAL	PERSONAL	VENTURE
A	11.26%	9.32%	12.83%	11.52%	10.54%	5.68%
B	18.78%	15.74%	21.62%	17.00%	18.20%	9.06%
C	22.77%	17.93%	27.35%	21.04%	18.97%	19.55%
D	92.81%	35.21%	54.00%	87.94%	43.55%	40.10%
E	100.00%	43.24%	51.75%	95.81%	48.30%	51.98%
F	100.00%	58.70%	65.63%	96.15%	50.00%	36.84%
G	100.00%	100.00%	100.00%	100.00%	100.00%	92.86%

Default Distribution by Home Ownership Type

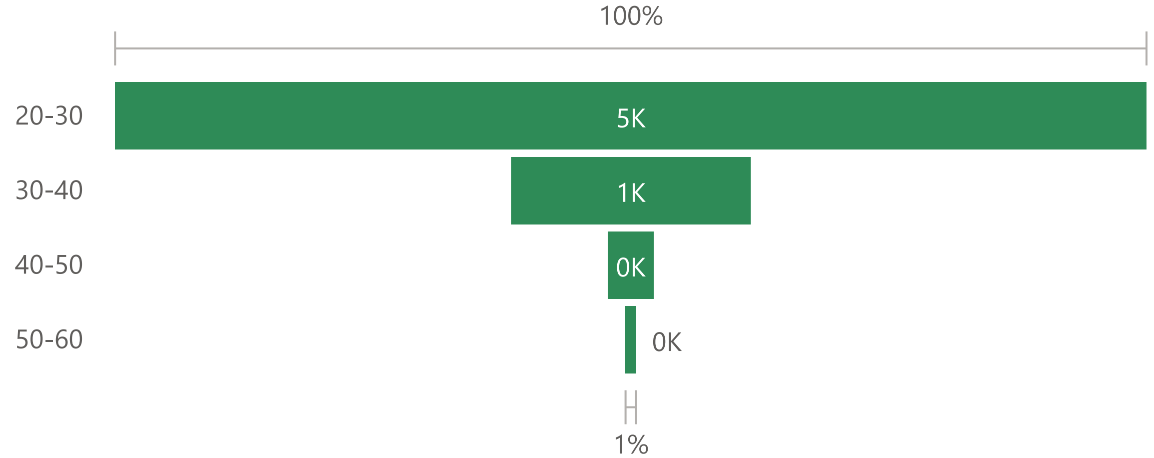
Loan Purpose Filter

All

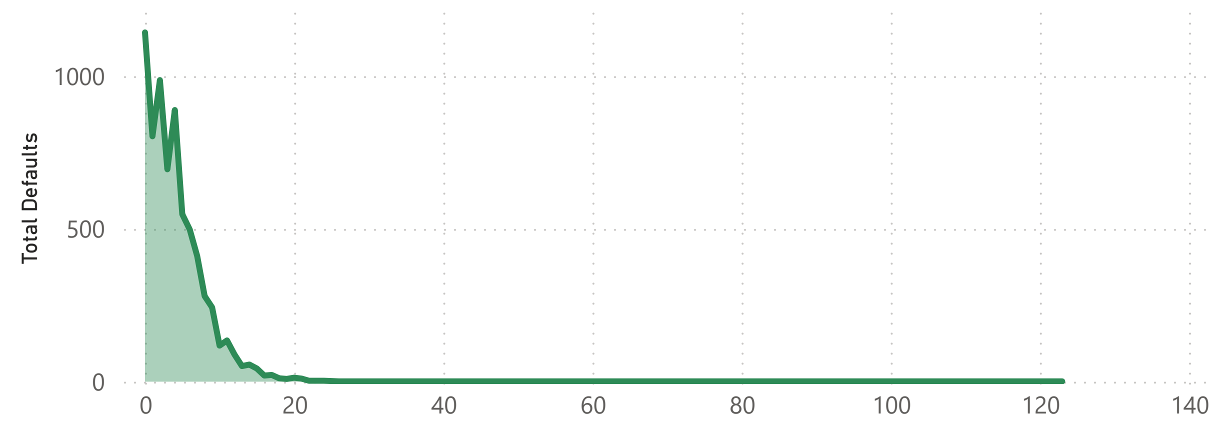
Age Group Filter

All

Age Group Default Analysis



Employment Length Impact on Loan Defaults



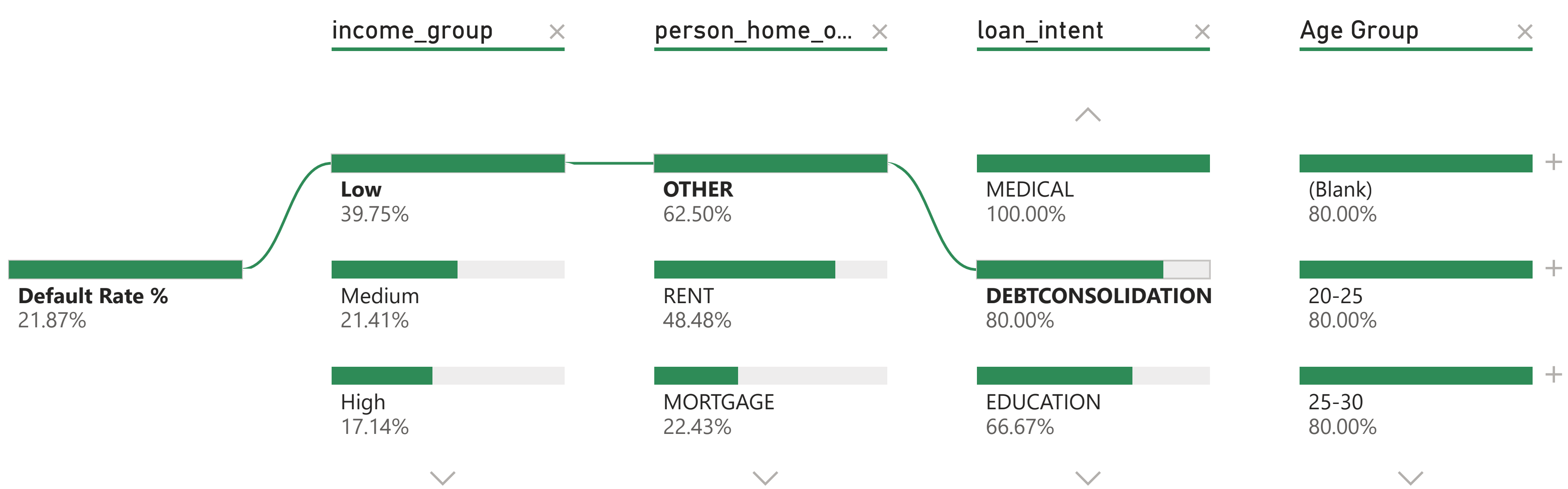
Loan Grade Filter

All

Home Ownership Filter

All

Drivers of Loan Default Risk by Customer Attributes



Key Insights

- **Loan Grade G exhibits the highest default rates across all loan purposes**, whereas Grade A borrowers consistently show the lowest risk, highlighting the strong relationship between credit grade and loan performance.
- **Younger borrowers (20–30 years)** account for the highest number of defaults, indicating that default risk decreases with age.
- **Customers living in rented properties represent the largest share of defaulters**, while homeowners and mortgage holders demonstrate relatively better repayment behavior.
- **Loan defaults are inversely related to employment length**. Borrowers with shorter employment histories are more likely to default, whereas default frequency declines as employment tenure increases.
- **Low-income customers exhibit the highest default risk**, suggesting that income level is one of the most significant drivers of loan repayment behavior.

Loan Grade Filter

All

PREDICTIVE ANALYTICS DASHBOARD

Best Accuracy
94.64%

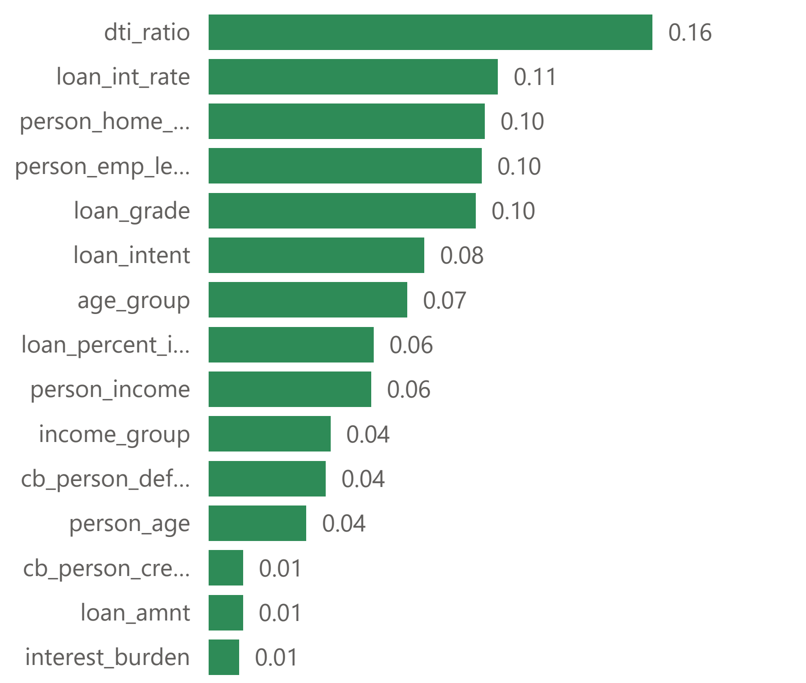
Best Precision
98.13%

Best Recall
99.98%

Best ROC AUC
94.65%

Best F1 Score
94.46%

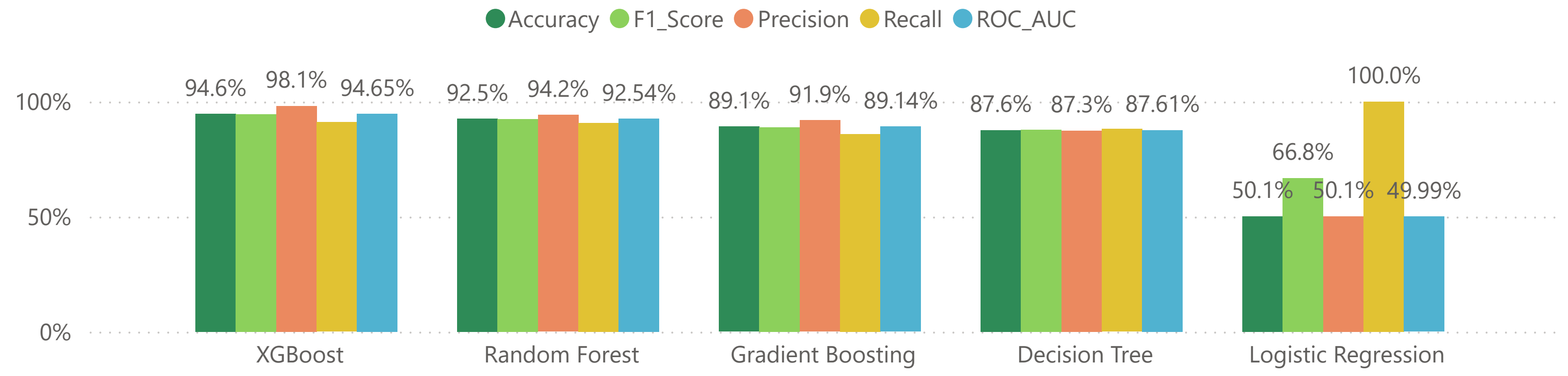
Top Predictors of Loan Default



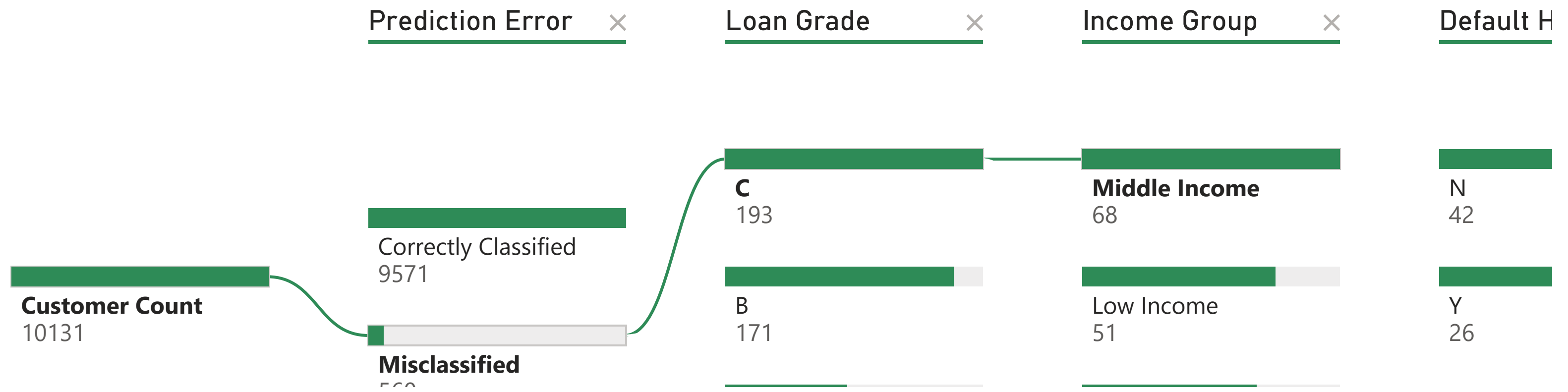
Feature Impact on Default Predictions

Feature	SHAP_Importance
age_group	0.25
cb_person_cred_hist_length	0.12
cb_person_default_on_file	0.07
Cluster	0.29
dti_ratio	1.03
income_group	0.22
interest_burden	0.08
loan_amnt	0.08
loan_grade	0.36
loan_int_rate	0.62
loan_intent	0.54
loan_percent_income	0.49
person_age	0.27
person_emp_length	1.14
person_home_ownership	0.58
person_income	0.66

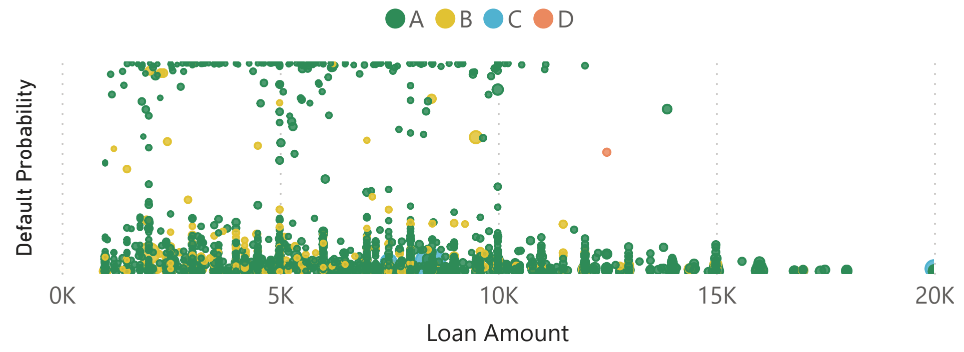
Comparison of Machine Learning Models



Drivers of Loan Default Risk by Customer Attributes



High-Risk Borrowers Distribution



Actual vs Predicted Loan Classification Flow

Pred/Act	Default	No Default
No Default	471	4962
Default	4609	89

Key Insights

- **XGBoost emerged as the best-performing model**, achieving **94.64% Accuracy, 98.13% Precision, 99.98% Recall, 94.65% ROC-AUC, and 94.46% F1-Score**, outperforming all other machine learning models.
- **Prediction errors are relatively low**, with most misclassifications occurring among **middle-income borrowers, mid-grade loans, and customers with both default and non-default histories**, indicating overlapping risk characteristics in these segments.
- The **Actual vs Predicted classification flow** shows that the model correctly classified the vast majority of customers, demonstrating strong predictive capability and reliability.
- According to **SHAP analysis**, the most

Dashboard

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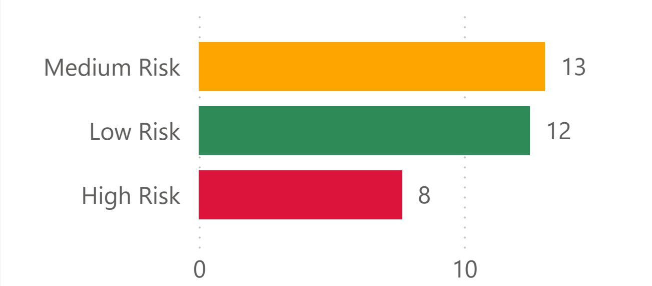
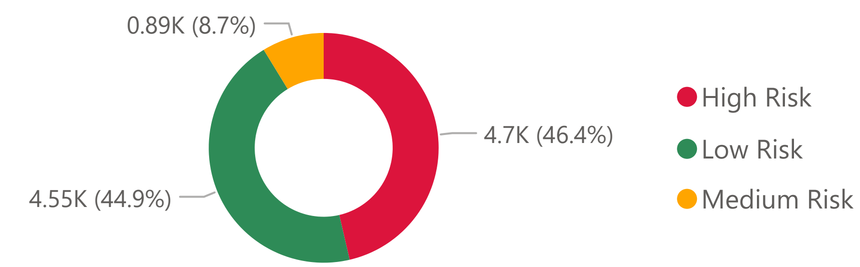
CUSTOMER SEGMENTATION AND RISK PROFILING

Employment Length by Cluster

Interest Rate by Cluster



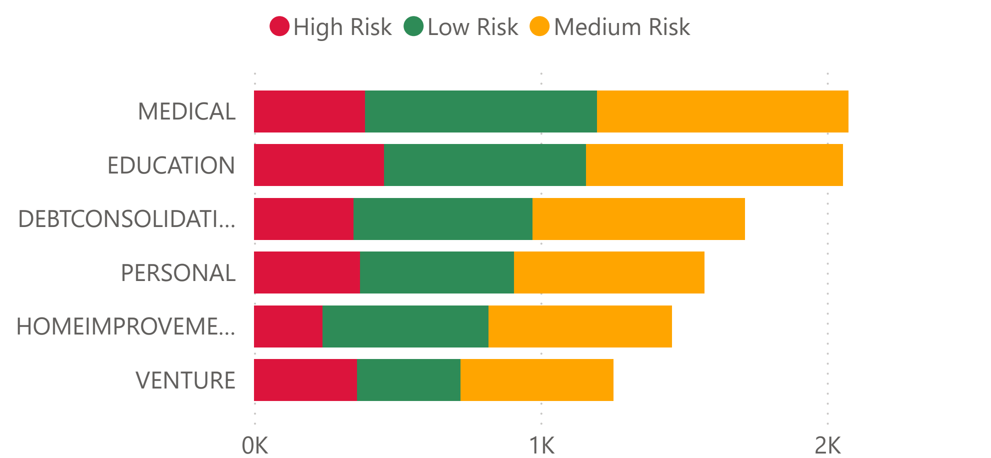
Risk Level Distribution



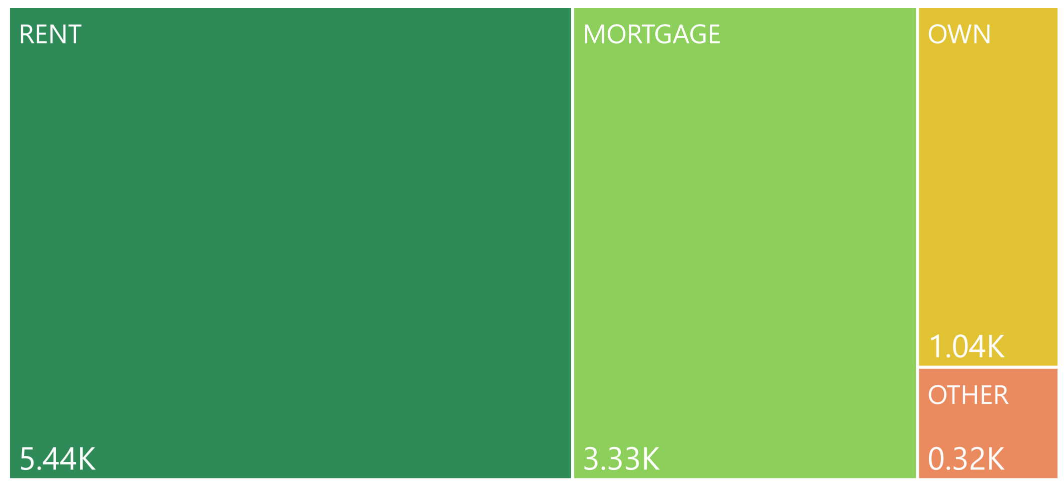
Risk Level Filter

- [ ] High Risk
- [ ] Low Risk
- [ ] Medium Risk

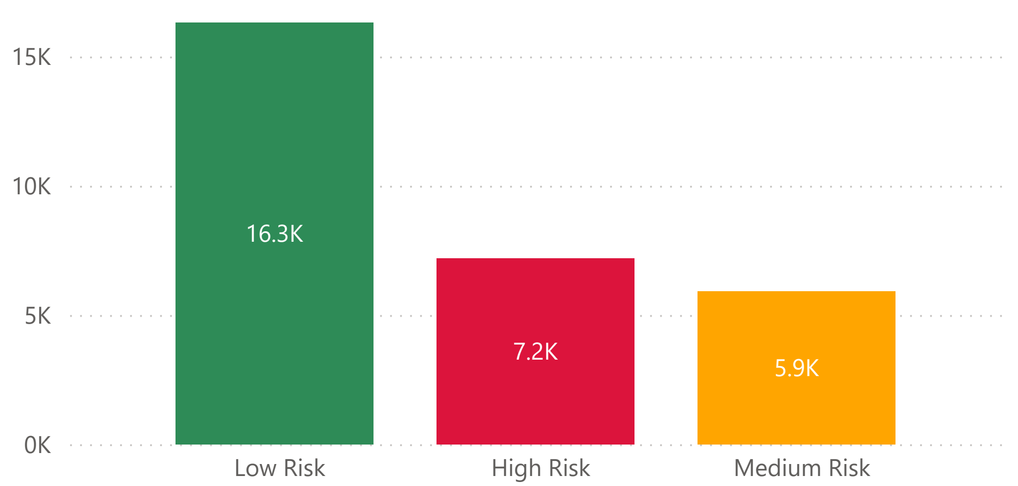
Loan Intent by Cluster



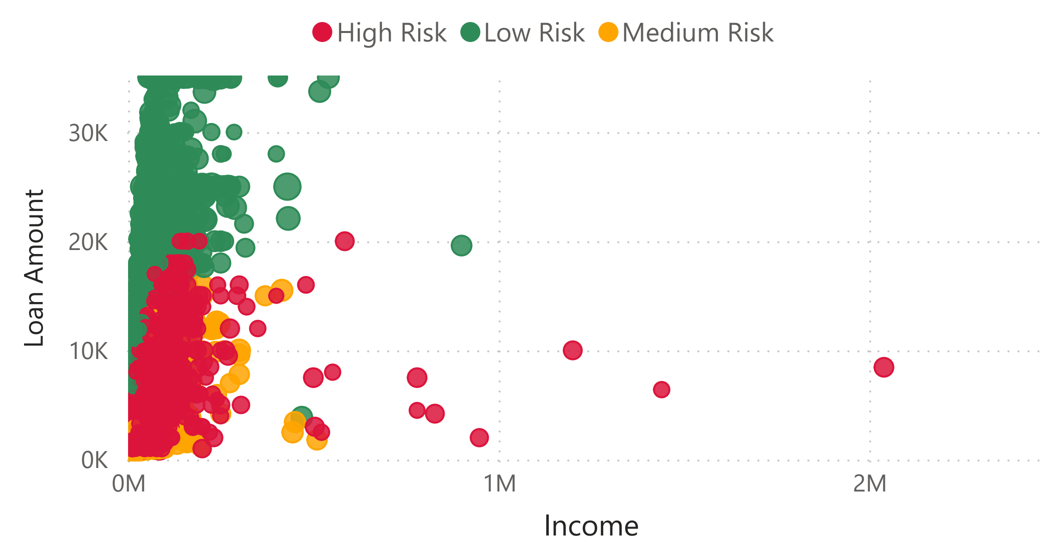
Home Ownership Type



Loan Amount by Cluster



Income vs Loan Amount by Risk Level



Key Insights

- **High-risk customers constitute the largest segment of the portfolio (46.4%)**, followed closely by low-risk customers (44.9%), indicating a relatively high concentration of risky borrowers.
- **Education and Medical loans exhibit the highest concentration of high-risk borrowers**, suggesting that these loan categories require closer monitoring and stricter underwriting policies.
- **High-risk customers generally have longer employment histories**, indicating that employment length alone is not sufficient to assess creditworthiness and should be evaluated alongside income and debt burden.
- **Rental customers account for the largest share of borrowers**, followed by mortgage holders, implying that housing status plays a significant role in customer segmentation.
- **Low-risk customers account for the highest total loan amount (₹16.3K)**, whereas high-risk and medium-risk customers contribute significantly smaller loan volumes, indicating that larger loan exposures are concentrated among safer borrowers.
- The **Income vs Loan Amount analysis reveals that most high-risk customers are concentrated in lower income and lower loan amount ranges**, while customers with higher incomes tend to belong to the low-risk segment.
- **Average interest rates are relatively similar**

Executive Summary and Recommendation

Recommendations

- Strengthen credit underwriting standards for high-risk borrowers, particularly those with low income, high debt-to-income ratios, and lower credit grades.
- Closely monitor Education and Medical loan segments, which exhibit relatively higher risk.
- Implement an early warning system using the XGBoost model to proactively identify customers with a high probability of default.
- Adopt risk-based lending strategies by incorporating customer income, housing status, employment history, and credit characteristics into loan approval decisions.
- Focus on expanding lending to low-risk customer segments to improve portfolio quality and profitability.
- Establish a long-term objective of reducing the portfolio default rate from **21.9% to below 5%**.

Future Scope

- Deploy the model using **Streamlit or Flask** for real-time loan default prediction.
- Integrate external data sources such as credit bureau scores and transaction histories to improve model performance.
- Develop real-time monitoring and early warning systems for high-risk borrowers.
- Explore advanced deep learning techniques to enhance predictive accuracy.
- Implement automated risk-based pricing and lending strategies.
- Extend the framework to support related applications such as fraud detection and customer lifetime value analysis.

Conclusion

This project successfully developed an **Advanced Credit Risk Analytics and Loan Default Prediction System** using machine learning, data mining, predictive analytics, and customer segmentation techniques. The analysis identified key drivers of loan default and provided actionable insights to improve credit risk management. Among all models evaluated, **XGBoost emerged as the best-performing model**, achieving **94.64% accuracy** and **94.65% ROC-AUC**, demonstrating strong predictive capability. Customer segmentation and SHAP analysis further enhanced the understanding of borrower behavior and risk profiles, enabling more informed and data-driven lending decisions.

- Overall, the project demonstrates how advanced analytics can help financial institutions minimize credit losses, improve portfolio health, and strengthen risk management practices.

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